PS3 · Karnaugh maps (Ch 4) — Worked Solutions

Source: Digital Logic homework for Sep. 18 · study-slides (vetame21.github.io/study-slides)

> [ADDED] The original problem sets have no solutions. These worked solutions were written for this site; every conversion and every Boolean result was checked in code (truth tables and an exact K-map minimizer), and the VHDL was compiled. Compare them with your lecturer's model answers when those are released.

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1 Use Karnaugh maps to find a minimal sum of products.

(a) $F = \Sigma_{x,y,z}(1, 3, 5, 6, 7)$

> **Answer:** $F = z + xy$

Cells 1, 3, 5, 7 all have $z = 1$: one group of 4 $\rightarrow z$. Cell 6 (110) pairs with 7 (111) $\rightarrow xy$. Two groups cover every 1.

Why this method: Make each group as large as possible (a power of 2), and use as few groups as possible.

(b) $F = \Sigma_{w,x,y,z}(1, 4, 5, 6, 7, 9, 14, 15)$

> **Answer:** $F = w'x + xy + x'y'z$

4, 5, 6, 7 (row $w'x$) $\rightarrow w'x$. 6, 7, 14, 15 (column xy) $\rightarrow xy$. 1 (0001) and 9 (1001) differ only in $w \rightarrow x'y'z$. The exact minimizer confirms this is the only minimal sum (3 terms, 7 literals).

Why this method: Look for the groups of 4 first; then cover the leftovers with the largest group that contains them.

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2 Use Karnaugh maps to find a minimal sum of products; d(…) are don't-cares.

(a) $F = \Sigma_{w,x,y,z}(0, 1, 2, 8, 11) + d(3, 9, 15)$

> **Answer:** $F = x'y' + w'x' + x'z$

$0, 1, 8, 9_d \rightarrow x'y'$ (group of 4 using $d9$). $0, 1, 2, 3_d \rightarrow w'x'$. $1, 3_d, 9_d, 11 \rightarrow x'z$ (the only way to put 11 in a group of 4). $d15$ is not needed.

Why this method: Use a don't-care only when it makes a group bigger; never cover a group of only d's.

(b) $F = \Sigma_{A,B,C,D}(1, 5, 12, 13, 14, 15) + d(7, 9)$

> **Answer:** $F = AB + C'D$

12, 13, 14, 15 $\rightarrow AB$. $1, 5, 9_d, 13 \rightarrow C'D$ (using $d9$). $d7$ is left as 0.

Why this method: Don't-cares let two groups of 4 replace smaller ones.